Abstract

On February 6, 2023, a devastating earthquake with a magnitude of 7.8 struck southern Turkey and northern Syria, with its epicenter near the city of Gaziantep. The seismic event caused extensive destruction, leaving thousands dead, displacing tens of thousands more, and crippling essential infrastructure throughout the region. The magnitude of the disaster underscored the urgent necessity for advanced emergency management systems that can rapidly identify vulnerable areas and guide efficient response strategies. This paper applies Geographic Information Systems (GIS) to evaluate the spatial distribution of risk factors associated with the 2023 earthquake. The study focuses on identifying high-risk seismic zones, assessing population vulnerability, and generating optimized evacuation plans for affected areas. Multiple datasets, including seismic hazard maps, population density statistics, building integrity data, transportation networks, and locations of emergency services, were integrated and analyzed using a combination of overlay techniques, network analysis, and suitability modeling. The results reveal critical vulnerabilities in densely populated areas with aging infrastructure and poor access to emergency routes. Based on the spatial patterns and outputs of the GIS analyses, the study proposes a set of data-driven recommendations to guide urban planners, emergency management officials, and policy makers. These include reevaluation of existing evacuation protocols, identification of optimal locations for emergency shelters, and suggestions for long-term urban resilience strategies. By presenting a comprehensive case study of GIS applications in earthquake disaster response, this research contributes to the growing body of literature advocating for spatially-informed crisis management policies.

Introduction

Natural disasters present some of the most significant challenges to modern society. Among these, earthquakes are particularly deadly due to their sudden onset and potential to cause massive damage to both lives and infrastructure. Unlike storms or floods, earthquakes typically strike without warning, leaving limited time for preemptive action. In regions such as Turkey, which sits atop several active fault lines, the risk of seismic events is an ever-present reality. The February 6, 2023, earthquake that devastated Gaziantep and surrounding areas was a stark reminder of this vulnerability. With a magnitude of 7.8, the quake was one of the strongest to hit the region in decades and resulted in widespread destruction. Buildings crumbled, critical infrastructure such as roads and bridges failed, and essential services were overwhelmed. The magnitude of the tragedy demanded not only immediate humanitarian relief but also a long-term reassessment of disaster preparedness and urban planning practices in seismically active areas.

Gaziantep, a major economic and cultural hub in southeastern Turkey, has seen rapid urban expansion in recent decades. Its population, now exceeding two million, resides in a mix of modern developments and older, often poorly constructed buildings. While new construction in Turkey has increasingly incorporated seismic-resistant design principles, older structures remain prevalent in many districts. Additionally, informal settlements—often developed without regulatory oversight—pose particular challenges during emergencies due to their density, lack of infrastructure, and reduced access to emergency services. These factors combine to increase the city's exposure and sensitivity to seismic events.

One of the most promising tools for improving disaster response is Geographic Information Systems (GIS). GIS allows for the integration and analysis of spatial data across multiple domains, making it possible to visualize hazards, assess risk exposure, and plan for effective mitigation. In the aftermath of an earthquake, GIS can be used to identify collapsed structures, locate populations in need, and model transportation routes for aid delivery and evacuation. When used proactively, GIS can also inform the development of safer urban designs, guide the location of critical facilities, and support training exercises and public awareness campaigns. This paper explores how GIS technologies can be leveraged to support earthquake response and planning in Gaziantep. It focuses on three core objectives: mapping high-risk zones, identifying vulnerable populations, and designing efficient evacuation and sheltering strategies. Through a series of spatial analyses, this research aims to demonstrate how GIS can support informed decision-making and contribute to the development of more resilient urban environments.

Background

Turkey is situated at the convergence of the Eurasian, Arabian, and African tectonic plates, placing the country within one of the most seismically active zones on Earth. The North Anatolian Fault (NAF) and East Anatolian Fault (EAF) are two of the most prominent fault lines traversing the country. Historically, both have produced numerous devastating earthquakes. For example, the 1939 Erzincan earthquake and the 1999 İzmit earthquake are among the most catastrophic events in Turkey's history, causing massive casualties and economic damage. These historical events have shaped Turkey's approach to earthquake management, leading to the development of new building codes and the establishment of agencies such as the Disaster and Emergency Management Authority (AFAD).

Despite these efforts, challenges persist, particularly in rapidly urbanizing regions like Gaziantep. The city's population growth has outpaced the development of resilient infrastructure. Many neighborhoods were built during periods when building regulations were either less stringent or poorly enforced. This issue is especially acute in informal settlements, where residents often construct homes without the guidance of structural engineers or the oversight of municipal authorities. These areas are typically characterized by narrow roads, limited open space, and inadequate access to emergency services—factors that significantly complicate disaster response efforts.

GIS has emerged as a powerful tool in disaster risk reduction and emergency management. It has been used extensively in countries such as Japan, the United States, and Indonesia to support real-time monitoring, hazard mapping, and post-disaster assessments. In the aftermath of the 2010 Haiti earthquake, for example, international aid organizations relied heavily on GIS to map damaged areas, identify priority regions for intervention, and coordinate logistics. Similarly, in the 2015 Nepal earthquake, GIS helped authorities identify isolated communities in need of aid and allocate resources more efficiently. GIS has also been used in Turkey for flood risk assessments, urban growth modeling, and environmental monitoring, but its potential for earthquake response remains underutilized.

Research has shown that effective disaster response requires a multi-faceted understanding of risk that includes not just the physical characteristics of the hazard but also the social, economic, and infrastructural vulnerabilities of the affected population. GIS enables the integration of diverse datasets—seismic hazard maps, demographic statistics, infrastructure inventories—into a unified analytical framework. This allows decision-makers to identify areas of compound risk, where multiple vulnerabilities intersect, and to prioritize interventions accordingly. By applying this approach to the 2023 Gaziantep earthquake, this study aims to contribute to a more holistic understanding of disaster preparedness in seismically active urban areas.

Methods

The methodology for this study was structured around the collection, integration, and analysis of multiple spatial datasets relevant to earthquake risk and response. These included seismic hazard maps, population distribution statistics, building infrastructure data, transportation networks, and the locations of critical facilities such as hospitals, fire stations, and emergency shelters. Data sources included the Kandilli Observatory and Earthquake Research Institute, AFAD, the Turkish Statistical Institute (TUIK), OpenStreetMap, and municipal records from the city of Gaziantep.

All datasets were converted to a common coordinate reference system to ensure consistency during analysis. Data cleaning involved standardizing attribute fields, resolving topological errors, and removing duplicate entries. Building footprint data were cross-verified with satellite imagery and cadastral records to improve accuracy. Where discrepancies existed, priority was given to official government sources. The resulting dataset served as the foundation for a series of spatial analyses conducted using ArcGIS Online.

The first analytical task was to perform an overlay analysis combining seismic hazard zones with population density and building age information. This analysis was designed to identify areas where ground shaking intensity was likely to coincide with high concentrations of people and structurally vulnerable buildings. Particular attention was paid to informal settlements, which were mapped using a combination of remote sensing and field survey data. These areas were flagged as high-risk zones requiring special attention in evacuation planning and resource allocation.

The second analytical task focused on evacuation modeling. A network dataset was constructed from the transportation network, incorporating both primary and secondary roads. Travel impedance values were assigned based on road type, width, surface condition, and assumed traffic congestion during emergencies. Network analysis tools were used to calculate the fastest evacuation routes from high-risk residential zones to nearby emergency shelters. The analysis also simulated road closures resulting from earthquake damage, enabling the identification of alternative routes and traffic bottlenecks.

The third task involved suitability analysis to determine the optimal locations for new emergency shelters. This analysis considered factors such as distance from fault lines, proximity to population centers, accessibility via major roads, availability of open space, and the presence of basic utilities. A weighted overlay method was employed, assigning scores to each criterion based on expert consultation and literature review. Areas with the highest composite scores were proposed as potential sites for temporary shelters. These were cross-referenced with municipal land use plans to ensure feasibility.

Visualizations were produced for each stage of the analysis, including hazard maps, vulnerability heatmaps, evacuation flowcharts, and suitability rankings. These maps were used to communicate findings to local authorities and to support scenario planning exercises. The methods employed in this study reflect best practices in spatial analysis and were informed by case studies from previous earthquake response efforts in Turkey and abroad.

Discussion

The results of this GIS-based analysis underscore the complex interplay between seismic hazards, urban infrastructure, and population vulnerability in Gaziantep. One of the most striking findings from the overlay analysis was the concentration of risk in the Sahinbey and Sehitkamil districts, which together house a substantial portion of the city's population. These areas contain many buildings constructed before 2000, when Turkey's seismic codes were updated following the 1999 İzmit earthquake. These structures often lack the necessary reinforcements to withstand major tremors, and many are multi-story apartment complexes, increasing the risk of catastrophic collapse. Coupled with high population density, this structural vulnerability creates conditions where a significant number of casualties and injuries are likely if another major earthquake strikes.

Further compounding the risk in these districts is the lack of sufficient open space for evacuation and emergency sheltering. In field visits conducted prior to the 2023 earthquake, urban planners noted the scarcity of parks or public plazas that could serve as safe assembly areas. This issue was confirmed by the spatial data, which showed that most green or vacant spaces were located far from the highest-risk residential zones. In an emergency, the inability of people to gather safely in open areas increases the likelihood of crowding, panic, and injury. The lack of centralized emergency gathering points also complicates rescue efforts, as emergency responders struggle to locate survivors and distribute essential supplies efficiently. Spatial disorganization in emergency assembly procedures can delay response times and hinder the implementation of coordinated relief strategies.

The network analysis provided valuable insights into the functionality of the city's transportation system under duress. While primary roads such as the D850 and D400 are wide and well-maintained, they are also surrounded by high-density development and are likely to be obstructed by debris following a major seismic event. This means that, although these roads might appear ideal for evacuation in theory, they may be impractical in the immediate aftermath of a disaster. Secondary roads offer some potential relief, but they are often narrower and not designed for high-capacity traffic. Additionally, some of these roads traverse older neighborhoods where building collapses could quickly block egress. The model highlighted several bottlenecks where congestion is likely to impede emergency response vehicles, especially in the first few hours following a quake.

Evacuation modeling also revealed spatial inequities in access to emergency services. Wealthier districts in the city's northwest had multiple routes leading to hospitals, fire stations, and shelters. By contrast, low-income neighborhoods in the southeast had limited access to such facilities, and what access did exist was often via a single, vulnerable route. This finding raises important questions about equity in disaster preparedness and the distribution of public resources. The city's emergency planning must go beyond general accessibility and include targeted investment in underserved areas to ensure that no community is disproportionately affected in a crisis. Developing community-based response hubs in these areas could be one such solution, providing localized first aid, emergency supplies, and communication tools during the critical early hours of a disaster.

The suitability analysis for emergency shelter placement yielded several promising options that had not been previously designated by the municipality. For example, a university campus in the city’s southwest quadrant was identified as highly suitable due to its large open areas, existing infrastructure (e.g., dormitories, cafeterias, restrooms), and proximity to high-risk residential zones. Similarly, a large sports complex near the airport was deemed appropriate for temporary sheltering because of its ample parking space, multiple buildings, and logistical accessibility. The analysis also suggested that vacant land near the city’s northern industrial district could be rapidly converted into a tent city if needed, provided water and sanitation infrastructure is installed. However, these spatial suggestions must be matched with political and institutional will. Land use planning in Gaziantep remains fragmented, and coordination between municipal departments is often lacking. Incorporating the outputs of GIS suitability models into official land use and zoning plans will be crucial to operationalize these findings.

Community input would be a valuable addition to these technical analyses. Residents often possess intimate knowledge of their neighborhoods, including informal paths, community centers, and gathering points that are not captured in official datasets. Future work could incorporate participatory GIS techniques, wherein community members contribute spatial data and validate model outputs. This approach has been successful in other disaster-prone regions, including Indonesia, the Philippines, and parts of Latin America, and has improved both the accuracy and legitimacy of planning efforts. Engaging local populations also enhances community trust and ensures that response strategies align with the lived realities of residents. Grassroots data collection methods, such as participatory mapping workshops or mobile app-based surveys, could be leveraged to deepen understanding of localized vulnerabilities.

Moreover, the integration of educational institutions, civil society organizations, and local businesses into the emergency planning process can build broader resilience. Universities, for instance, can host simulation exercises and generate scenario models, while non-profits can assist in public education campaigns. Local retailers and logistics providers could be incorporated into supply chain planning, ensuring that food, water, and medical aid reach affected zones swiftly. GIS can serve as the central coordination platform for these diverse actors, creating a unified spatial database that enables collaboration and resource sharing.

Overall, the findings from this research highlight the urgent need for comprehensive, spatially-informed disaster planning in Gaziantep. The city’s current emergency response plans do not appear to adequately account for the spatial distribution of risk, nor do they leverage the full potential of GIS technology. By integrating GIS into all phases of the disaster management cycle—mitigation, preparedness, response, and recovery—Gaziantep can enhance its resilience to future seismic events. Moreover, the methodological framework developed in this study is replicable and can be adapted for use in other Turkish cities or in international contexts facing similar challenges. The study's approach of combining hazard overlays, population vulnerability analysis, and evacuation modeling provides a clear template for decision-makers seeking evidence-based planning tools. While no single model can fully predict the complexity of a disaster, spatial data analysis enables cities to prepare more intelligently, allocate resources more equitably, and respond more rapidly when lives are on the line.

Conclusion

The 2023 earthquake that struck Gaziantep and its surrounding regions was a tragic reminder of the destructive potential of seismic events, especially in densely populated urban environments. In the aftermath of such a disaster, questions inevitably arise regarding the adequacy of existing preparedness measures and the tools available to mitigate risk and manage response. This research demonstrates that Geographic Information Systems (GIS) offer a robust and versatile solution for addressing the multifaceted challenges posed by earthquakes.

Through a comprehensive analysis of spatial datasets—including seismic hazard maps, population density distributions, building characteristics, and transportation networks—this study identified key zones of vulnerability within Gaziantep. The findings revealed a concerning overlap between high-risk seismic zones and areas with structurally vulnerable buildings and limited access to emergency services. Network analysis underscored the fragility of the city's evacuation routes, particularly in older neighborhoods with constrained road systems. Suitability analysis provided actionable insights for the optimal placement of emergency shelters, highlighting previously overlooked sites with strong logistical potential.

This research contributes not only to academic understanding of GIS in disaster management but also offers practical recommendations for municipal decision-makers. These include updating evacuation routes based on real-time traffic simulations, retrofitting structurally weak buildings in high-risk zones, prioritizing shelter development in underserved districts, and incorporating participatory mapping to capture local knowledge. In addition, the city should consider establishing a centralized GIS unit within its disaster management office to ensure that spatial data are continuously updated and readily accessible during emergencies. Such a unit could also serve as a liaison with national agencies and international partners, facilitating knowledge exchange and technology transfer.

Looking forward, future work could expand upon this study by integrating real-time data streams from remote sensing platforms, mobile apps, and Internet of Things (IoT) sensors to enable dynamic monitoring of infrastructure and population movements during crises. Machine learning models could be trained on historical disaster data to predict likely damage zones and assist in resource allocation. Furthermore, collaborations with universities and international NGOs could facilitate the transfer of technical expertise and funding necessary to implement these strategies. Beyond technology, emphasis must also be placed on community resilience through education, training, and inclusive planning.

In conclusion, the application of GIS to the Gaziantep earthquake case study has demonstrated the enormous potential of spatial analysis in supporting disaster preparedness, response, and recovery. GIS enables stakeholders to visualize risk, identify vulnerable populations, plan strategic interventions, and optimize the allocation of resources. By embracing GIS as a core component of its emergency management strategy, Gaziantep—and other cities like it—can move toward a future in which natural disasters, though inevitable, do not necessarily result in widespread tragedy. The use of GIS transforms passive vulnerability into active readiness, giving cities the power to anticipate, adapt, and act before, during, and after disasters. As climate change and urbanization continue to increase the frequency and intensity of natural hazards, adopting GIS-based strategies is no longer optional—it is imperative for survival and resilience.

Maps

1. **Seismic Hazard Update for Gaziantep (2023)**
   This map provides detailed seismic hazard information for the Gaziantep region, highlighting fault lines and areas of potential seismic activity.

2. **[Gaziantep District Population Statistics]**
   Detailed population data for Gaziantep's districts, including Şahinbey and Şehitkamil, which are among the most densely populated.

References

AFAD (Disaster and Emergency Management Presidency). (2018, March 30). *Turkey's new earthquake hazard map is published*. Disaster and Emergency Management Authority.https://en.afad.gov.tr/turkeys-new-earthquake-hazard-map-is-published

Amiran, A., Behnam, B., & Seyedin, S. (2024). AI-based model for site-selecting earthquake emergency shelters. *Scientific Reports, 14*(1), 29033.https://doi.org/10.1038/s41598-024-80586-w

Ganjehi, S., & Norouzi Khatiri, K. (2021). Determination of emergency roads to emergency accommodation using loss analysis results. *Geoenvironmental Disasters, 8*(1), 15.https://doi.org/10.1186/s40677-021-00190-2

Kandilli Observatory and Earthquake Research Institute. (2023, February 6). *06 February 2023 Mw 7.7 Sofalaca-Şehitkamil (Gaziantep) earthquake*. Boğaziçi University.http://www.krdae.boun.edu.tr/sismo/2/06-subat-2023-ml7-4-sofalaca-sehitkamil-gaziantep-depremi/

OpenStreetMap contributors. (2023). *OpenStreetMap data extracts*.https://www.openstreetmap.org

Soltani, A., Ardalan, A., Akbarisari, A., Rashed, A., & Javdan, M. (2014). Site selection criteria for sheltering after earthquakes: A systematic review. *PLOS Currents: Disasters*, 6.https://doi.org/10.1371/currents.dis.17ad1f98fb85be80785d0a81ced6a7a6

Turkish Statistical Institute (TÜİK). (2023). *Address-based population registration system (ADNKS) results, 31 December 2022*.https://data.tuik.gov.tr